GME Analysis — Hubbard L=6 Ne=6 t=1.0 U=4.0 — n=3 (mixed)

```
System:  L=6  Ne=6  t=1.0  U=4.0  mode=uniform
RDM:  n=3  sites=(0, 1, 2)  dim=8x8
Tr(rho) = 1.00000000   Tr(rho^2) = 0.35932201   rank = 8

============================================================
  LOWER BOUNDS
============================================================


  C_LB  notes eqs.(6-9)       = 0.00000000
    best z1 = [0, 0, 0]
    best z2 = [1, 1, 1]

  C_LB  best of all               = 0.00000000

============================================================
  CONVEX ROOF
============================================================

  C_num (existing numerical)       = 0.02249929
  C_GME (unitary optimization)     = 0.17299221
    restarts = 200   optimizer = L-BFGS-B

  Decomposition from unitary optimization:
    i=0: p=0.014324  C_GME=0.022108  contrib=0.000317
    i=1: p=0.188408  C_GME=0.272727  contrib=0.051384
    i=2: p=0.147746  C_GME=0.073168  contrib=0.010810
    i=3: p=0.140344  C_GME=0.039041  contrib=0.005479
    i=4: p=0.110404  C_GME=0.219914  contrib=0.024279
    i=5: p=0.133208  C_GME=0.193424  contrib=0.025766
    i=6: p=0.135432  C_GME=0.190059  contrib=0.025740
    i=7: p=0.130134  C_GME=0.224515  contrib=0.029217

============================================================
  C_LB=0.000000 <= C_GME=0.172992 <= C_UB=0.627165
```

GME Analysis — Hubbard L=6 Ne=6 t=1.0 U=4.0 — n=4 (pure)

```
System:  L=6  Ne=6  t=1.0  U=4.0  mode=uniform
RDM:  n=4  sites=(0, 1, 2, 3)  dim=16x16
Tr(rho) = 1.00000000   Tr(rho^2) = 0.39958921   rank = 11

  C_GME exact (eq.1) = 1.00000000

=========================================================
  LOWER BOUNDS
=========================================================


  C_LB  notes eqs.(6-9)       = 0.05288569
    best z1 = [0, 1, 0, 1]
    best z2 = [1, 0, 1, 0]

  C_LB  best of all               = 0.05288569
  Gap  C_exact - C_LB_best        = 0.94711431

=========================================================
  CONVEX ROOF  (pure state)
=========================================================

  C_num (existing numerical)      = 0.17000514
  C_GME (unitary optimization)    = 0.26763203
  C_exact (eq.1)                  = 1.00000000

=========================================================
  C_LB=0.052886 <= C_GME=0.267632 <= C_UB=0.627165
  C_exact = 1.000000
```